

```
vp = ; %P-wave velocity (m/s)
vs = ; %S-wave velocity (m/s)
rho = 2300; %density (kg/m^3)

mu = vs^2*rho*10^-9 %Shear Modulus (GPa)
lambda = (vp^2*rho-2*mu)*10^-9 %Lame's Lambda (GPa)
K = (lambda+2*mu/3) %bulk modulus (GPa)
E = (9*K*mu)/(3*K+mu) %Young's Modulus (GPa)
nu = lambda/(2*(lambda+mu)) %Poisson's Ratio
```